

Total marks – 70**QUESTION 1-10 Objective-response questions.****(10marks)****Write answers on the Objective response answers booklet****Marks****Multiple Choice**

1. Which one of the following is NOT a polynomial:

1

- (A) $(\sqrt{3}x - 1)^3$ (B) $\frac{x + 3}{5}$
- (C) $\frac{1}{x^2} + 4x + 7$ (D) $-2x^3 + 0.5x^2$

2. The value of k if $x + 1$ is a factor of $x^3 - 2x^2 - kx + 5$ is:

1

- (A) 4 (B) 2
- (C) -2 (D) -6

3. How many solutions would $\cos^2 \theta = \frac{1}{4}$ have in the domain $0 \leq \theta \leq 2\pi$?

1

- (A) 0. (B) 2 (C) 1 (D) 4

4. Which of the following gives the correct simplification of $\sin x \sec x$.

1

- (A) 1. (B) $\cot x$ (C) $\tan x$ (D) $\operatorname{cosec} x$

5. Find the gradient of the normal to the parabola $x^2 = 4ay$ at $(2at, at^2)$.

1

- (A) t (B) $-\frac{1}{t}$ (C) $\frac{1}{t}$ (D) $-t$

6. The exact value of $\tan^{-1}\left(\tan\frac{2\pi}{3}\right)$ is 1

(A) $-\frac{\pi}{3}$ (B) $-\sqrt{3}$ (C) $-\frac{2\pi}{3}$ (D) $\frac{2\pi}{3}$

7. Which of the following is the correct expansion for $\cos\left(\theta - \frac{\pi}{3}\right)$ 1

(A) $\frac{1}{2}(\cos\theta - \sqrt{3}\sin\theta)$ (B) $\cos\theta - \frac{1}{2}$
(C) $\frac{1}{2}\cos\theta$ (D) $\frac{1}{2}(\cos\theta + \sqrt{3}\sin\theta)$

8. Which of the following is the correct general solution for $\sin x = \frac{1}{2}$. 1

(A) $x = n\pi + (-1)^n \frac{\pi}{6}$ (B) $x = n\pi + \frac{\pi}{6}$
(C) $x = 2n\pi + \frac{\pi}{6}$ (D) $x = 2n\pi \pm \frac{\pi}{6}$

9. Which of the following statements is FALSE. 1

(A) $\cos^{-1}(-\theta) = -\cos^{-1}\theta$. (B) $\sin^{-1}(-\theta) = -\sin^{-1}\theta$
(C) $\tan^{-1}(-\theta) = -\tan^{-1}\theta$ (D) $\cos^{-1}(-\theta) = \pi - \cos^{-1}\theta$

10. Consider the function $f(x) = x^2 - 6x$. Which of the following gives the correct domain of $f(x)$ for which there exists an inverse function, $f^{-1}(x)$. 1

(A) All real x . (B) $0 \leq x \leq 6$
(C) $x \leq 3$ or $x \geq 3$ (D) $x \geq 0$

End of Question Multiple Choice

QUESTION 11 (15 marks) **START A NEW BOOKLET****Mark**

- (a) Find the acute angle between the lines: $x - 2y + 1 = 0$ and $y = 5x - 4$, to the nearest degree. **3**

- (b) Solve the inequality $\frac{4}{x+1} \leq 3$ and graph your solution on the number line. **3**

- (c) Find the coordinates of the point P which divides the interval joining A(-4, 3) and B(-8, -9) externally in the ratio 2 : 3. **2**

- (d) Use the method of mathematical induction to prove that

$$1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1) \text{ for integer } n \geq 1$$

4

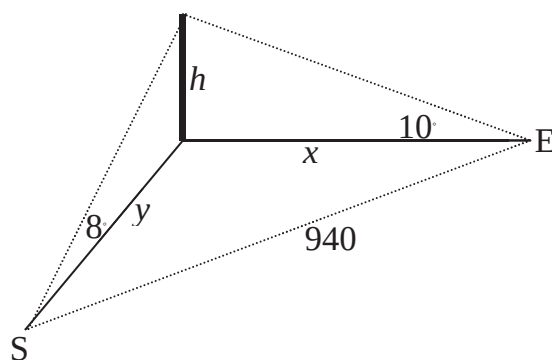
- (e) Find the cartesian equation of the curve represented by the following parametric equations;

$$\begin{aligned} x &= 5 \sin \theta \\ y &= 5 \cos \theta \end{aligned}$$

3**End of Question 11**

QUESTION 12 (15 marks) **START A NEW BOOKLET****Marks**

- (a) (i) Write $\sin x - \sqrt{3} \cos x$ in the form $R \sin(x - \alpha)$, where $R > 0$ and $0 \leq x \leq \frac{\pi}{2}$. 1
- (ii) Hence, or otherwise solve the equation $\sin x - \sqrt{3} \cos x = 1$, $0 \leq x \leq 2\pi$. 2
- (b) Evaluate $\int_0^{\frac{\pi}{4}} \cos^2(2x) dx$. 2
- (c) Using the substitution $t = \tan \frac{\theta}{2}$,
- (i) Write down an expression for $\tan \theta$, in terms of t . 1
- (ii) Use this expression to show $\tan 15^\circ = 2 - \sqrt{3}$ 2
- (d) Prove $\frac{\sin 3A}{\sin A} + \frac{\cos 3A}{\cos A} = 4 \cos 2A$. 3
- (e) A surveyor who is y metres south of a tower sees the top of it with an angle of elevation 8° . A second surveyor is x metres east of the tower. From his position the angle of elevation is 10° to the top of the tower. The two surveyors are 940m apart.
- (i) Show that $y = h \cot 8^\circ$ 1
- (ii) Find the height of the tower to the nearest metre. 3

**End of Question 12**

QUESTION 13 (15 marks) **START A NEW BOOKLET****Marks**

(a) Find the exact values of each of the following, showing all working.

(i) $\cos^{-1}\left(\sin\frac{\pi}{3}\right)$ 1

(ii) $\sin\left(2\tan^{-1}\frac{1}{3}\right)$. 2

(b) Find $\frac{d}{dx}(x^2 \cos^{-1} 2x)$ 2

(c) Find $\int \frac{-1}{\sqrt{1-4x^2}} dx$ 2

(d) Evaluate $\int_0^1 \frac{5}{3+x^2} dx$ 2

(e) (i) Find the domain and range for the function $y = 5\sin^{-1}\left(\frac{x}{\pi}\right)$. 2

(ii) Sketch $y = 5\sin^{-1}\left(\frac{x}{\pi}\right)$. 2

(f) For what values of x does $y = \frac{xe^x}{2}$ have an inverse function. Use differentiation to explain your answer. 2

End of Question 13

QUESTION 14 (15 marks) **START A NEW BOOKLET** **Marks**

- (a) The polynomial $p(x) = x^3 + ax + b$ has $(x - 5)$ as one of its factors and has a remainder of -60 when divided by $(x + 5)$. Find the values of a and b . **3**
- (b) The polynomial equation $x^3 - 5x^2 + 7x + 5 = 0$ has 3 roots, α, β, γ
- (i) Find $\alpha + \beta + \gamma$ **1**
- (ii) Find $\alpha\beta + \beta\gamma + \gamma\alpha$ **1**
- (iii) Find $\alpha^2 + \beta^2 + \gamma^2$ **2**
- (c) Given that $x^3 - 3x^2 + 2x - 4 = (x^2 + 2) \times Q(x) + R(x)$. **3**
- Find $Q(x)$ and $R(x)$.
- (d) Sketch the following polynomial. Clearly show all intercepts.
- $$P(x) = x(x + 2)^2(1 - x)^3.$$
- 2**
- (e) Solve the equation $3x^3 - 17x^2 - 8x + 12 = 0$ given that the product of two of the roots is 4. **3**

END OF THE PAPER

Student Number: _____ **ANSWERS** _____

Questions 1-10 Multiple Choice Answer Sheet

- | | | | | |
|-----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1. | A ☐ | B ☐ | C ☒ | D ☐ |
| 2. | A ☐ | B ☐ | C ☒ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☒ |
| 4. | A ☐ | B ☐ | C ☒ | D ☐ |
| 5. | A ☐ | B ☒ | C ☐ | D ☐ |
| 6. | A ☒ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☒ |
| 8. | A ☒ | B ☐ | C ☐ | D ☐ |
| 9. | A ☒ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☒ | D ☐ |

11(a)

$$x - 2y + 1 = 0 \quad \therefore m_1 = \frac{1}{2} \quad \text{for } y = 5x - 4 \quad \therefore m_2 = 5 \quad \checkmark$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\therefore \tan \theta = \left| \frac{\frac{1}{2} - 5}{1 + \frac{5}{2}} \right| = \left| \frac{-9}{7} \right| \quad \checkmark$$

$$\therefore \theta = 52^\circ \quad (\text{nearest deg}) \quad \checkmark$$

11(b)

$$\frac{4}{x+1} \leq 3$$

$$\therefore 4(x+1) \leq 3(x+1)^2$$

$$\therefore 3(x+1)^2 - 4(x+1) \geq 0$$

$$(x+1)[3x+3-4] \geq 0$$

$$(x+1)(3x-1) \geq 0$$

$$\therefore x \geq \frac{1}{3} \quad \text{or} \quad x < -1 \quad \checkmark \checkmark \checkmark$$

11(c) A(-4, 3) and B(-8, -9)

Externally so use 2 : -3

$$x = \frac{-3(-4) + 2(-8)}{2 + (-3)} \quad y = \frac{-3(3) + 2(-9)}{2 + (-3)}$$

$$= \frac{12 - 16}{-1} \quad = \frac{-9 - 18}{-1}$$

$$= \frac{-4}{-1} \quad = \frac{-27}{-1}$$

$$= 4 \quad = 27 \quad \checkmark \checkmark$$

P is the point (4, 27)

11(d)

$$1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1)$$

$$\text{Test } n = 1. \quad 3^{1-1} = \frac{1}{2}(3^1 - 1) = 1 \quad \therefore \text{true for } n = 1 \quad \checkmark$$

$$n = k \text{ i.e. } 1 + 3 + 9 + \dots + 3^{k-1} = \frac{1}{2}(3^k - 1)$$

Prove true for $n = k + 1$

$$\begin{aligned} \text{i.e. } 1 + 3 + 9 + \dots + 3^{k-1} + 3^k &= \frac{1}{2}(3^k - 1) + 3^{(k+1)-1} & \checkmark \\ &= \frac{1}{2}(3^k - 1) + 3^k \\ &= \frac{1}{2}(3^k - 1 + 2 \cdot 3^k) \\ &= \frac{1}{2}(3 \cdot 3^k - 1) \\ &= \frac{1}{2}(3^{k+1} - 1) \end{aligned}$$

$$\text{Which is in the form } \frac{1}{2}(3^n - 1) \text{ where } n = k + 1 \quad \checkmark$$

$\therefore$ True for $n = k + 1$ when true for $n = k$,

But it is true for $n = 1 \therefore$ True for $n = 1 + 1 = 2$

And $2 + 1 = 3$ Etc.

Hence by Mathematical Induction $\checkmark$ (accept any conclusion)

$$1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1)$$

11(e)

$$x = 5 \sin \theta$$

$$y = 5 \cos \theta$$

$$\therefore x^2 = 25 \sin^2 \theta \quad \checkmark$$

$$y^2 = 25 \cos^2 \theta$$

$$\therefore x^2 + y^2 = 25(\sin^2 \theta + \cos^2 \theta) \quad \checkmark$$

$$\therefore x^2 + y^2 = 25 \quad \checkmark$$

QUESTION 12 (15 marks) **START A NEW BOOKLET****Marks**

(a) (i) If $\sin x - \sqrt{3} \cos x = R \sin(x - \alpha)$ then

$$R = \sqrt{1^2 + (\sqrt{3})^2} \text{ and } \tan \alpha = \sqrt{3}$$

$$\therefore R = 2 \quad \text{and} \quad \alpha = \frac{\pi}{3}$$

$$\text{So } \sin x - \sqrt{3} \cos x = 2 \sin\left(x - \frac{\pi}{3}\right)$$

1

(ii) Hence, or otherwise solve the equation $\sin x - \sqrt{3} \cos x = 1$, $0 \leq x \leq 2\pi$.

$$\sin x - \sqrt{3} \cos x = 1$$

$$2 \sin\left(x - \frac{\pi}{3}\right) = 1$$

$$\sin\left(x - \frac{\pi}{3}\right) = \frac{1}{2}$$

$$x - \frac{\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$x = \frac{\pi}{2}, \frac{7\pi}{6}$$

$$12(b) \quad \int_0^{\frac{\pi}{4}} \cos^2(2x) dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} (1 + \cos(4x)) \cos^2 dx \quad \checkmark$$

$$= \frac{1}{2} \left[\left[x + \frac{1}{4} \sin(4x) \right] \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \left(\frac{\pi}{4} + 0 - 0 \right)$$

$$= \frac{\pi}{8} \quad \checkmark$$

$$12(c) \quad (i) \quad t = \tan \frac{\theta}{2}$$

$$\therefore \tan \theta = \frac{2t}{1-t^2} \quad \checkmark$$

$$(ii) \quad \therefore \tan 30^\circ = \frac{2t}{1-t^2} \quad \checkmark$$

$$\therefore 1-t^2 = 2\sqrt{3}t$$

$$\therefore t^2 + 2\sqrt{3}t - 1 = 0$$

$$\therefore t = \frac{-2\sqrt{3} + \sqrt{12+4}}{2} \quad \checkmark$$

$$\therefore t = 2 - \sqrt{3}$$

,

$$12(d) \quad \frac{\sin 3A}{\sin A} + \frac{\cos 3A}{\cos A} = 4\cos 2A$$

$$LHS = \frac{\sin 3A}{\sin A} + \frac{\cos 3A}{\cos A}$$

$$= \frac{\sin 3A \cos A + \cos 3A \sin A}{\sin A \cos A} \quad \checkmark$$

$$= \frac{\sin(3A + A)}{\sin A \cos A}$$

$$= \frac{\sin 4A}{\frac{1}{2}\sin 2A} \quad \checkmark$$

$$= \frac{4\sin 2A \cos 2A}{\sin 2A} \quad \checkmark$$

$$= 4\cos 2A = RHS$$

.

$$12(e) \quad (i) \quad \frac{h}{y} = \tan 8^\circ \quad \text{so } \frac{y}{h} = \cot 8^\circ \quad \text{ie } y = h \cot 8^\circ$$

$$(ii) \quad \text{likewise } x = h \cot 10^\circ$$

$$\text{By Pythagoras } x^2 + y^2 = 940^2 \quad \text{i.e.}$$

$$h^2 \cot^2 8^\circ + h^2 \cot^2 10^\circ = 883600$$

$$h^2 (\cot^2 8^\circ + \cot^2 10^\circ) = 883600$$

$$h^2 = \frac{883600}{\cot^2 8^\circ + \cot^2 10^\circ} = \frac{883600}{82.7919} = 10672.54$$

$$h = 103 \text{ metres (to nearest metre)}$$

End of Question 12

QUESTION 13 (15 marks) **START A NEW BOOKLET**

Marks

(a) Find the exact values of

$$(i) \quad \cos^{-1}\left(\sin \frac{\pi}{3}\right) = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) \\ = \frac{\pi}{6} \quad \checkmark$$

$$(ii) \quad \sin\left[2 \tan^{-1} \frac{1}{3}\right] = 2 \sin \theta \cos \theta \quad \text{let } \theta = \tan^{-1} \frac{1}{3} \\ = 2 \times \frac{1}{\sqrt{10}} \times \frac{3}{\sqrt{10}} \quad \checkmark \\ = \frac{3}{5} \quad \checkmark$$

(b) Find $\frac{d(x^2 \cos^{-1} 2x)}{dx}$

$$\frac{d}{dx}(2x^2 \cos^{-1} 2x) = 2x^2 \frac{-1}{\sqrt{\frac{1}{4} - x^2}} + 4x \cos^{-1} 2x \\ = 2x^2 \frac{-2}{\sqrt{1 - 4x^2}} + 4x \cos^{-1} 2x \\ = \frac{-4x^2}{\sqrt{1 - 4x^2}} + 4x \cos^{-1} 2x$$

$\checkmark \checkmark$

$$13(c) \quad \int \frac{-1}{\sqrt{1 - 4x^2}} dx = \frac{1}{2} \int \frac{-1}{\sqrt{\frac{1}{4} - x^2}} dx$$

$$= \frac{1}{2} \cos^{-1}(2x) + c \quad \checkmark \checkmark \text{ accept answer only}$$

$$13(d) \int_0^1 \frac{5}{3+x^2} dx = \frac{5}{\sqrt{3}} \left[\tan^{-1} \left(\frac{x}{\sqrt{3}} \right) \right]_0^1 \quad \checkmark$$

$$= \frac{5}{\sqrt{3}} \left[\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) - 0 \right]$$

$$= \frac{5}{\sqrt{3}} \times \frac{\pi}{6}$$

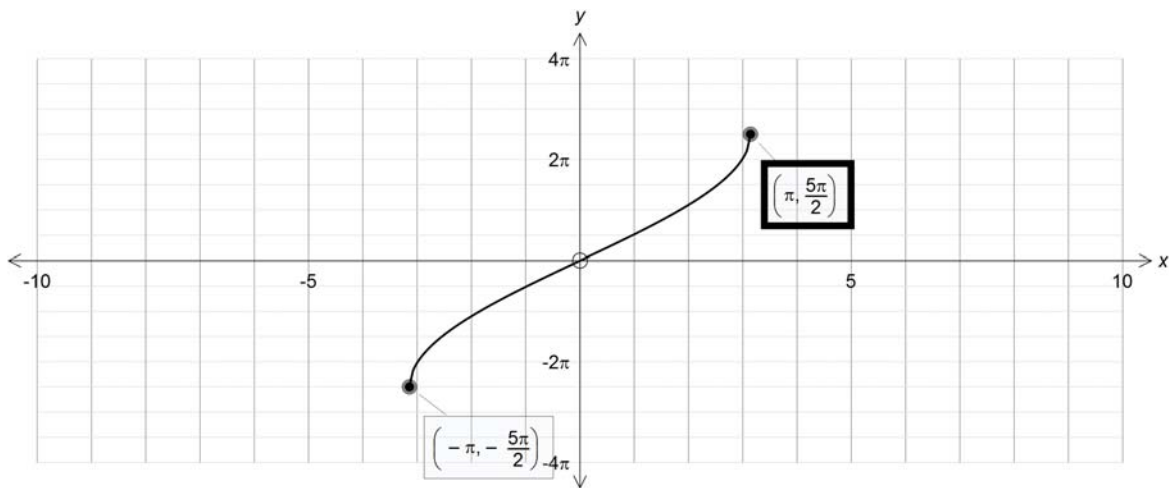
$$= \frac{5\pi}{6\sqrt{3}} = \frac{5\sqrt{3}\pi}{18} \quad \checkmark$$

(e) (i) For the function $y = 5\sin^{-1}\left(\frac{x}{\pi}\right)$.

Domain $-1 \leq \frac{x}{\pi} \leq 1$ so $-\pi \leq x \leq \pi$ $\checkmark$

Range $-\frac{5\pi}{2} \leq y \leq \frac{5\pi}{2}$ $\checkmark$ 2

(ii) . 2



- 13(f) For what values of x does $y = \frac{xe^x}{2}$ have an inverse function. Use differentiation to explain your answer.

$$y = \frac{xe^x}{2}$$

$$\frac{dy}{dx} = \frac{xe^x}{2} + \frac{e^x}{2} \quad \text{product rule}$$

$$= \frac{e^x}{2}(x + 1) \quad \checkmark$$

$$\therefore x \geq -1 \quad \text{or} \quad x \leq -1$$

$$\text{inverse exists when } \frac{dy}{dx} < 0 \text{ or } \frac{dy}{dx} > 0 \quad \checkmark$$

QUESTION 14 (15 marks) **START A NEW BOOKLET**

Marks

(a)

$$5a + b + 125 = 0 \quad \text{①}$$

$$-5a + b - 65 = 0 \quad \text{②}$$

$$2b + 60 = 0 \quad \text{①} + \text{②}$$

$$2b = -60$$

$$b = -30$$

$$5a - 30 + 125 = 0$$

$$5a = -95$$

$$a = -19$$

✓ ✓ ✓

(b) The polynomial equation $x^3 - 5x^2 + 7x + 5 = 0$ has 3 roots, α, β, γ

$$(i) \alpha + \beta + \gamma = -\frac{b}{a} = 5 \quad \checkmark$$

$$(ii) \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = 7 \quad \checkmark$$

$$(iii) (\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\beta\gamma + 2\gamma\alpha \quad \checkmark \checkmark$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = 5^2 - 2 \times 7 = 11$$

(c) Given that $x^3 - 3x^2 + 2x - 4 = (x^2 + 2) \times Q(x) + R(x)$.

3

Find $Q(x)$ and $R(x)$.

$x^3 - 3x^2 + 2x - 4$ is divided by $x^2 + 2$

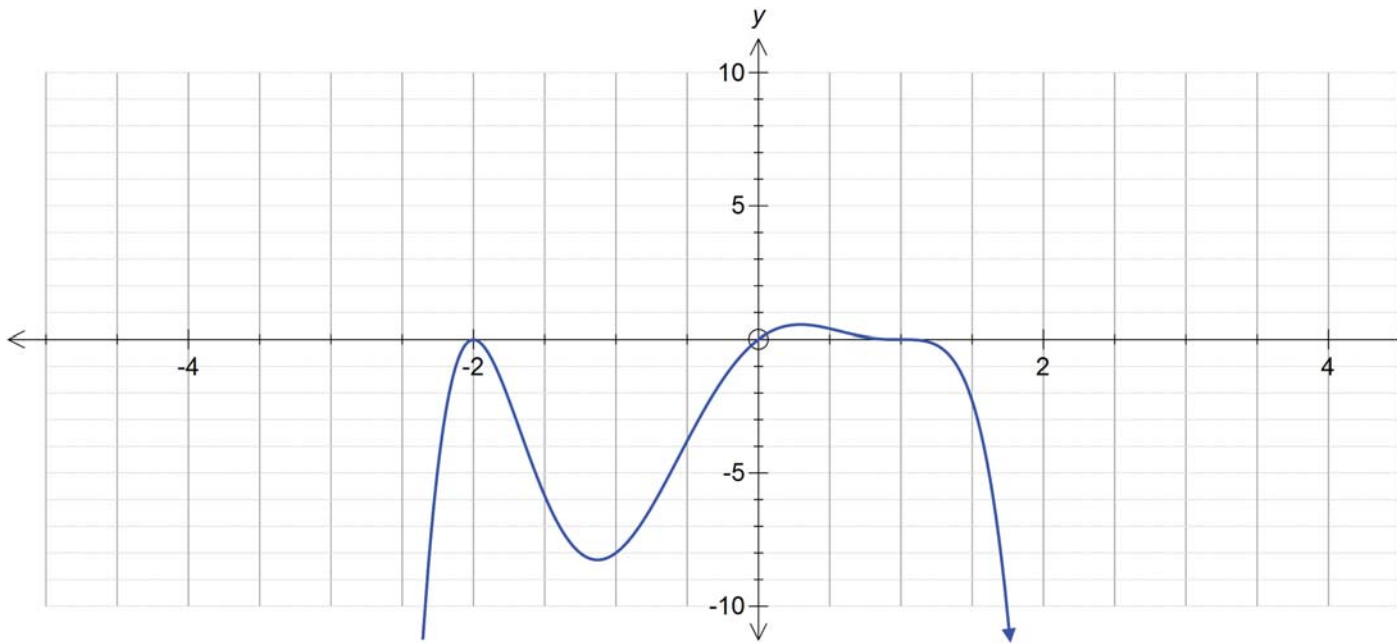
$$\begin{array}{r} x - 3 \\ x^2 + 2 \overline{) x^3 - 3x^2 + 2x - 4} \\ \underline{x^3 + 2x} \\ -3x^2 - 4 \\ \underline{-3x^2 - 6} \\ 2 \end{array}$$

$$Q(x) = x - 3 \text{ and } R(x) = 2 \quad \checkmark \checkmark \checkmark$$

(d) Sketch the following polynomial. Clearly show all intercepts.

$$P(x) = x(x+2)^2(1-x)^3.$$

✓✓



✓✓

14(e) let roots be $\alpha, \frac{4}{\alpha}, \beta$

ACCEPT OTHER METHODS.

$$\therefore \text{product } \alpha \times \frac{4}{\alpha} \times \beta = -\frac{12}{3}$$

$$\beta = -1$$

$$\therefore \text{sum } \alpha + \frac{4}{\alpha} + \beta = \frac{17}{3}$$

$$\alpha + \frac{4}{\alpha} - 1 = \frac{17}{3}$$

$$3\alpha^2 - 20\alpha + 12 = 0$$

$$(3\alpha - 2)(\alpha - 6) = 0$$

roots are $-1, \frac{2}{3}$ and 6 .

✓✓✓

END OF THE PAPER